Project Test Plan

ACE
CYBER R&D Program

PI: Teaundra Thompson

University: Charleston Southern University

Project Title: IDS Resiliency Against Evasion Attacks

1. Test Plan Purpose and Best Practices:

This template is structured to be clear, rigorous, and promote reproducibility. It contains placeholders '[in brackets]' that you should replace with your project's specific information. The explanatory notes in italics are for your guidance and should be removed from the final document.

IDS Resiliency Against Evasion Attacks Test Plan

Document ID:	2001
Document Title:	Test Plan v1.5
Project Title:	IDS Resiliency Against Evasion Attacks
Version #:	1.5
Status:	Ready to implement
Author(s):	Teaundra Thompson
Date:	2025-12-17

1.0 Objective

This document outlines the test plan for evaluating a baseline MLP machine learning model-based IDS against evasion attacks.

The primary objective of this test plan is to execute a series of controlled experiments to collect data necessary to validate or invalidate the following hypothesis:

Hypothesis: If the NSL-KDD dataset is used for training a MLP model, the MLP will not recognize evasion attacks utilizing encapsulation, but it will recognize DoS attacks.

This plan will directly measure the relationship between the independent variable(s):NSL-KDD dataset, and the dependent variable(s): detection of attack types.

2.0 Scope

2.1 In-Scope

Testing of: Baseline IDS (Zeek v8.0.1) with MLP ML model component (Google Colab)

Execution of: *SYN Flood* DDoS attack using HPING3, brute-force SSH password cracking using HPING3 , overlapping fragmentation using python script, suspicious TCP connections from Spyware,.

Measurement of: Detection accuracy, true/false positive rate, and F-1 score

Testing on the specific environment defined in the Testbed Configuration Plan - TBCP-001

2.2 Out-of-Scope

- Novel evasion techniques
- Deployable IDS
- Testing of hardware
- ML/AI user interface

3.0 Test Environment

All testing will be conducted in accordance with the established Testbed and its defined configuration as noted in TBCP-001.

3.1 Key Components

Note: Adapt this section as necessary to fit your research project by changing the column name headers and editing the table.

Component	Host	Network Adapter
Zeek VM	VirtualBox	NAT and Host-Only
Kali Linux VM	VirtualBox	Host-Only
MLP Model	Google Colab	N/A (Cloud environment)

4.0 Prerequisites / Entry Criteria

The following conditions must be met before commencing any test case execution:

1. Zeek VM must be powered on, network connectivity is required
2. Open snapshot named "Begin Test Cases" (Start of test cases execution)
    a. Multiple test cases can be completed in one sitting
    b. In the case of needing to stop executions, power off the VM in a "save state" and resume from that point
3. Navigate to respective directory
4. Have Project Notebook open and ready for documentation of test case execution
5. Open Wireshark for packet capture

Note: I created all the pcap files for the testcases before implementing the testcase to allow for testing in a timely manner.

5.0 Test Cases

This is the core execution section where each test case is a distinct experiment.

Test Case ID: TC-001

Title: Baseline Measurement - No Attack

Objective: Determine the TPR and FPR when a pcap with normal traffic is inserted into IDS. Also ensure that the MLP model can understand Zeek's conn.log. **Test case is pass/fail**

Step-by-Step Procedure:

1. Preparation: Verify all Entry Criteria from Section 4.0 are met.
2. Open FireFox on vm and engage in light web browsing for 3 minutes
3. Save pcap file as "normalTraffic.pcapng" in TC1 folder
4. Via CLI, change directory to "testcases/TC1"
5. Feed pcap into zeek via CLI "zeek -r normalTraffic.pcap"
6. Upload conn.log into Colab environment (manually)
7. Preprocess the log
8. Analyze the log
9. Verify MLP output displays "No anomalies detected"
10. If necessary, power off vm in "save state"

Data to be Collected:

- normalTraffic.pcap from Wireshark
- conn.log from Zeek
- MLP analysis of Zeek log

Expected Outcomes / Success Criteria:

- Zeek successfully generates conn.log
- MLP accepts the conn.log
- MLP outputs "Normal Traffic"

Test Case ID: TC-002

Title: Attack Scenario – DDoS Attack / SYN Flood

Objective: Determine if the MLP model can detect a DDoS attack. Note that the UFW-ZeekData24 set does not contain a CSV file for the Impact Tactic. **Test case is exploratory.**

Step-by-Step Procedure:

1. Preparation: Verify all Entry Criteria from Section 4.0 are met.
    a. When starting capture on Zeek VM, capture on interface enp0s8
2. Boot up Kali Linux VM
3. Open Wireshark on Kali Linux VM and start capture on interface enp0n8 to cross check that packets are coming through
4. On Zeek VM, run command **sudo hping3 -S –flood –rand-source -p 80 192.168.56.102**
5. Let command run for 1 minute and 30 seconds, then kill it to prevent the host machine from crashing.
6. Verify the packets were coming thought on the Kali Linux VM by checking the Wireshark capture
    a. Discard pcap file on Kali Linux
    b. Shutdown Kali Linux VM
7. Save pcap file as **SYNflood.pcapng** (Zeek VM)
8. Via CLI, change directory to "testcases/TC2"
9. Feed pcap into zeek via CLI **zeek -r SYNFlood.pcapng**
10. Upload conn.log into Colab environment (manually)
11. Preprocess the log
12. Analyze the log
13. Screenshot MLP output

Data to be Collected:

- SYNFlood.pcapng from Wireshark (modified via Scapy)
- conn.log file from Zeek
- MLP analysis of Zeek log

Expected Outcomes / Success Criteria:

- Zeek successfully generates conn.log
- MLP accepts the conn.log

Test Case ID: TC-003

Title: Attack Scenario – Evasion using Overlapping Fragmentation

Objective: Determine the accuracy of the IDS when faced with typical evasion attacks. **Test case is exploratory.**

Step-by-Step Procedure:

1. Preparation: Verify all Entry Criteria from Section 4.0 are met.
2. Run python script created to simulate overlapping fragmented packets, **sudo python3 overlapFragment.py**
    a. Script was generated by Claude. Script sends fragmented overlapping packets in waves every 8-15 seconds over a course of 3 minutes.
    b. Used host machine for destination IP
3. Open FireFox on VM and engage in light web browsing
4. After three minutes stop capture and save pcap file as **fragmentation.pcapng**
5. Via CLI, run command **cd testcases/TC3**
6. Feed it into Zeek via CLI: **zeek -r fragmentation.pcapng**
7. Upload conn.log into Colab
8. Feed conn.log into MLP
9. Analyze MLP output
10. If necessary, power off vm in "save state"

Data to be Collected:

- **fragmentation.pcapng** from Wireshark
- conn.log file from Zeek
- MLP analysis of Zeek log

Expected Outcomes / Success Criteria:

- Zeek successfully generates conn.log
- MLP accepts the conn.log

Test Case ID: TC-004

Title: Attack Scenario – Spyware with Persistence

Objective: Determine if the IDS can detect malicious TCP connections made by spyware.

Step-by-Step Procedure:

1. Preparation: Verify all Entry Criteria from Section 4.0 are met.

2. Procure spyware.pcap from Recorded Future Triage
    a. Link for pcap: https://tria.ge/251122-d7a5macm6s/behavioral1
3. Feed it into Zeek with command **zeek -r spyware.pcap**
4. Upload conn.log into Colab environment
5. Feed conn.log into MLP
6. Analyze MLP output
7. If necessary, power off VM in "save state"

Data to be Collected:

- **defensiveSpyware.pcap**
- conn.log file from Zeek
- MLP analysis of Zeek log

Expected Outcomes / Success Criteria:

- Zeek successfully generates conn.log
- MLP accepts the conn.log

Test Case ID: TC-005

Title: Attack Scenario – Evasive Password Cracking

Objective: Determine if the MLP can detect password cracking attempts that are spread out over a period of time. Password cracking falls under the Credential Access tactic, which the MLP is trained on. **This test case is exploratory.**

Step-by-Step Procedure:

1. Preparation: Verify all Entry Criteria from Section 4.0 are met.
2. Run command "sudo hping3 -p 22 -i 2 10.0.2.2"
    a. Targeting host machine's SSH (port 22) and sending a packet every 2 seconds
3. Open FireFox on VM and engage in light web browsing
4. After three minutes stop capture and save pcap file as **bruteForce.pcapng**
5. Via CLI, change directory with command **cd testcases/TC5**
6. Feed the pcap into Zeek with command **zeek -r bruteForce.pcapng**
7. Upload conn.log into Colab environment (manually)
8. Preprocess the log
9. Analyze the log
10. Document MLP output

11. If necessary, power off vm in “save state”

Data to be Collected:

- bruteForce.pcapng
- conn.log file from Zeek
- MLP analysis of Zeek log

Expected Outcomes / Success Criteria:

- Zeek successfully generates conn.log
- MLP accepts the conn.log

Test Case ID: TC-006

Title: Attack Scenario – Network Scanning

Objective: Determine if the MLP can detect port scanning ; consider the fact the MLP was trained on the Reconnaissance tactic. **Test case is pass/fail**

Step-by-Step Procedure:

1. Preparation: Verify all Entry Criteria from Section 4.0 are met.
2. Download **SS_Scan.pcapg** from Nmap-and-Wireshark-Lab GitHub repository
    a. https://github.com/Ryan-Sapone/Nmap-and-Wireshark-Lab/blob/main/SS%20Scan.pcapng
3. Via CLI, run command **cd Desktop/testcases/TC6**
4. Feed the pcap into Zeek with command **zeek -r SS_Scan.pcapng**
5. Upload conn.log into Colab environment (manually)
6. Preprocess the log
7. Analyze the log
8. Document MLP output

Data to be Collected:

- SS_Scan.pcapng
- conn.log file from Zeek
- MLP analysis of Zeek log

Expected Outcomes / Success Criteria:

- Zeek successfully generates conn.log

- MLP accepts the conn.log
- MLP detects reconnaissance tactic (pass test case)

Test Case ID: TC-007

Title: Attack Scenario – Evasive Network Scanning

Objective: Determine if the MLP can detect slow fragmented port scanning; **Test case is observatory**

Step-by-Step Procedure:

1. Preparation: Verify all Entry Criteria from Section 4.0 are met.
    a. In Wireshark on Zeek VM, capturing packets on interfaces enpos3 and enpos8
2. Power on Kali Linux VM
3. Run command **ifconfig** on Zeek VM to get IP address
4. On Kali Linux VM open terminal and run command **sudo nmap -sS -T1 -f <Zeek VM IP address>** for three minutes
5. While the command is running engage in light web browsing on Zeek VM
6. Stop the packet capture after three minutes and kill the command
7. Save the pcap as **Slow_Scan.pcapng** in **testCases/TC7**
8. In Zeek VM run command (via CLI) **cd Desktop/testCases/TC7**
9. Feed pcap into Zeek (via CLI) **zeek -r Slow_Scan.pcapng**
10. Upload conn.log into Colab environment (manually)
11. Preprocess the log
12. Analyze the log
13. Document MLP output

Data to be Collected:

- Slow_Scan.pcapng
- conn.log file from Zeek
- MLP analysis of Zeek log

Expected Outcomes / Success Criteria:

- Zeek successfully generates conn.log
- MLP accepts the conn.log
- MLP provides analysis

7.0 Roles & Responsibilities

Role	Name	Responsibility
Test Executor	T. Thompson	Executing the test cases as defined and logging results
Reviewer	T. Thompson	Reviewing and approving the Test Plan for execution.
Data Collection	T. Thompson	Collecting test results and analyzing the data

8.0 References

ID #	Document Title	Location / Link
1	Project Charter	https://csuniv0-my.sharepoint.com/personal/trthompson_student_csuniv_edu/Documents/2025-Fall-ACE_Cyber_Project_Charter%20v2.0.docx
2	Literature Map	2025-Fall-ACE_Cyber_Project_Preliminary_Literature_Map_v2.0
3	Research Question & Hypothesis	2025-Fall-ACE_Cyber_Project_Research-Hypothesis_v.1.3
4	Research Design	https://csuniv0-my.sharepoint.com/personal/trthompson_student_csuniv_edu/Documents/ACE-Cyber-Project-Research-Design-Template.docx
5	Testbed setup & Config Plan	https://csuniv0-my.sharepoint.com/personal/trthompson_student_csuniv_edu/Documents/ACE_Cyber_Project%20Testbed%20Setup%20&%20Configuration%20Template%20v1.4.docx
6	Data Acquisition Plan	https://csuniv0-my.sharepoint.com/personal/trthompson_student_csuniv_edu/Documents/ACE_Cyber_Project_Data-Acq&Sythentic-Data-Gen-Report_v.1.1.docx
7	Research Project Notebook	https://csuniv0-my.sharepoint.com/personal/trthompson_student_csuniv_edu/Documents/8%20ACE-Cyber%20Project%20Research%20Notebook%20TEMPLATE-v1.docx